

## Exercise 1:

```
usermaleesha@Maleesha:~/Testing/OpenMP_Lab2$ mpicc -o program program.c
usermaleesha@Maleesha:~/Testing/OpenMP_Lab2$ mpirun -np 4 ./program
Root filled array with values 1 to 1000000
Rank 0: summed indices [0, 250000) => local_sum = 31250125000
Rank 1: summed indices [250000, 500000) => local_sum = 93750125000

[Bcast] Total sum      = 5000000500000
[Bcast] Expected      = 5000000500000
[Bcast] Correct?      = YES
[Bcast] Time          = 0.0313 sec
Rank 2: summed indices [500000, 750000) => local_sum = 156250125000
Rank 3: summed indices [750000, 1000000) => local_sum = 218750125000
```

## Exercise 2:

```
usermaleesha@Maleesha:~/Testing/OpenMP_Lab2$ mpicc sum_scatter.c -o sum_scatter
usermaleesha@Maleesha:~/Testing/OpenMP_Lab2$ mpirun -np 4 ./sum_scatter
Root filled array with values 1 to 1000000
Rank 2: local_sum = 156250125000
Rank 3: local_sum = 218750125000
Rank 0: local_sum = 31250125000

[Scatter] Total sum = 5000000500000
[Scatter] Expected  = 5000000500000
[Scatter] Correct?   = YES
[Scatter] Time       = 0.0027 sec
Rank 1: local_sum = 93750125000
```

## Exercise 3:

```
usermaleesha@Maleesha:~/Testing/OpenMP_Lab2$ mpicc sum_gather.c -o sum_gather
usermaleesha@Maleesha:~/Testing/OpenMP_Lab2$ mpirun -np 4 ./sum_gather
Root filled array with values 1 to 1000000
Rank 0: local_sum = 31250125000

[Gather] Total sum = 5000000500000
[Gather] Expected  = 5000000500000
[Gather] Correct?   = YES
[Gather] Time       = 0.0017 sec
Rank 1: local_sum = 93750125000
Rank 2: local_sum = 156250125000
Rank 3: local_sum = 218750125000
```

#### Exercise 4:

```
usermaleesha@Maleesha:~/Testing/OpenMP_Lab2$ mpicc sum_reduce.c -o sum_reduce
usermaleesha@Maleesha:~/Testing/OpenMP_Lab2$ mpirun -np 4 ./sum_reduce
Root filled array with values 1 to 1000000
Rank 2: local_sum = 156250125000
Rank 3: local_sum = 218750125000
Rank 0: local_sum = 31250125000

[Reduce] Total sum = 500000500000
[Reduce] Expected   = 500000500000
[Reduce] Correct?   = YES
[Reduce] Time       = 0.0023 sec
Rank 1: local_sum = 93750125000
```

#### Exercise 5:

```
usermaleesha@Maleesha:~/Testing/OpenMP_Lab2$ mpicc sum_allreduce.c -o sum_allreduce
usermaleesha@Maleesha:~/Testing/OpenMP_Lab2$ mpirun -np 4 ./sum_allreduce
Root filled array with values 1 to 1000000
Rank 1: local_sum = 93750125000, total_sum = 500000500000, contribution = 18.75%
Rank 2: local_sum = 156250125000, total_sum = 500000500000, contribution = 31.25%
Rank 3: local_sum = 218750125000, total_sum = 500000500000, contribution = 43.75%
Rank 0: local_sum = 31250125000, total_sum = 500000500000, contribution = 6.25%

[Allreduce] Total sum = 500000500000
[Allreduce] Expected   = 500000500000
[Allreduce] Correct?   = YES
[Allreduce] Time       = 0.0014 sec
```

#### Exercise 6:

```
usermaleesha@Maleesha:~/Testing/OpenMP_Lab2$ mpicc sum_scan.c -o sum_scan
usermaleesha@Maleesha:~/Testing/OpenMP_Lab2$ mpirun -np 4 ./sum_scan
Root filled array with values 1 to 1000000
Rank 3: local_sum = 218750125000, prefix_sum = 500000500000, sum_before_me = 281250375000, expected_prefix = 500000500000
0, correct = YES

[Scan] Last rank prefix_sum = 500000500000
[Scan] Expected global sum = 500000500000
[Scan] Correct?           = YES
[Scan] Time               = 0.0043 sec
Rank 0: local_sum = 31250125000, prefix_sum = 31250125000, sum_before_me = 0, expected_prefix = 31250125000, correct = YES
Rank 1: local_sum = 93750125000, prefix_sum = 125000250000, sum_before_me = 31250125000, expected_prefix = 125000250000,
correct = YES
Rank 2: local_sum = 156250125000, prefix_sum = 281250375000, sum_before_me = 125000250000, expected_prefix = 281250375000
0, correct = YES
```